

SMART-Rain: iTransformer for Climate-Aware Multivariate Rainfall Forecasting

Abstract—Accurate rainfall forecasting is needed for climate monitoring, water resource management, and disaster mitigation in regions with limited ground observations. However, rainfall data exhibit nonlinearity, temporal variability, and skewed distributions, making prediction challenging. This paper proposes SMART-Rain framework for multivariate rainfall forecasting. The paper utilizes around four decades (1982–2025) of subnational rainfall observations and integrates temporal, anomaly, cyclical, and spatial embedding features to capture regional variability. Three sequence models, namely LSTM, Transformer Encoder, and iTransformer, are evaluated under a unified experimental setting with Bayesian hyperparameter optimization. The results of SMART-Rain framework show that Transformer-based models outperform recurrent architectures. The iTransformer achieves the best performance with an R^2 score of 0.76, demonstrating capability in learning long-range temporal dependencies and multivariate feature interactions. The SMART-Rain framework capture seasonal patterns and rainfall trends, predicting rainfall events remains challenging due to data imbalance and distribution skewness. The SMART-Rain findings confirm the effectiveness of attention for long-term rainfall forecasting and provide useful insights for climate-related time-series prediction.

Index Terms—Rainfall forecasting, time-series prediction, deep learning, Transformer, iTransformer, LSTM, spatio-temporal modeling, climate data analytics, multivariate forecasting

I. INTRODUCTION

Rainfall forecasting plays a vital role in climate monitoring, water resource management, agriculture, and disaster risk reduction under increasing climate variability and extreme weather events [1]–[3]. However, rainfall is complex due to its nonlinear, intermittent, and skewed nature, making it a challenging time-series forecasting problem. Recent advancements in satellite-based datasets have improved the availability of long-term rainfall observations. The Climate Hazards Group InfraRed Precipitation with Stations (CHIRPS) dataset provides a reliable, high-resolution precipitation record spanning more than four decades [4]. Such datasets enable the development of data-driven models capable of capturing long-term temporal patterns and regional variability, particularly in data-scarce regions.

Deep learning techniques have been adopted for rainfall prediction due to their ability to model complex temporal dependencies [5]. In precipitation predicting tasks, Long Short-Term Memory (LSTM) has shown encouraging results [6]. An LSTM-based hybrid model for precipitation forecasting was suggested by the authors of [7], demonstrating gains over conventional machine learning techniques. Similarly, The authors of [8] applied LSTM, GRU, and CNN models for extreme rainfall prediction using multi-source climate data,

highlighting the effectiveness of recurrent models in capturing temporal dependencies.

Despite their advantages, LSTM-based models often struggle with long-range dependencies and complex spatio-temporal interactions. Transformer has become a potent solution to the problem [9]–[11], using self-attention to identify global dependencies in time-series data [12], [13]. Transformers’ efficacy in precipitation forecasting has been shown in recent investigations. By creating a Transformer for precipitation nowcasting, the inventors of [14] were able to outperform traditional methods. Additionally, by combining Transformer modules with convolutional and recurrent networks, the developers of [15] improved predicting accuracy.

Recently, the iTransformer model has been introduced as an advanced architecture for multivariate time-series forecasting, where variables are treated as tokens to better capture inter-feature relationships [16]. The design has shown performance across multiple forecasting benchmarks, making it a promising candidate for rainfall prediction tasks involving complex multivariate inputs.

Nevertheless, some challenges remain in existing such as (i) many works focus on short-term forecasting or station-level data, limiting their ability to generalize across large spatial regions; (ii) extreme rainfall events are often underrepresented in training data, leading to biased predictions and poor performance in high-intensity scenarios; (iii) limited attention has been given to integrating spatial identifiers, such as subnational regions, into deep learning models to capture geographical variability. To address the limitations, this paper proposes SMART-Rain framework for multivariate rainfall forecasting using long-term CHIRPS-derived data at subnational levels. The SMART-Rain framework evaluates and compares LSTM, Transformer Encoder, and iTransformer models under a unified experimental setting, incorporating temporal, statistical, and spatial features. The contributions of this paper are summarized as follows:

- We propose SMART-Rain framework for multivariate rainfall forecasting. SMART-Rain integrates data pre-processing, spatio-temporal feature engineering, and sequence modeling. Rainfall dataset encompassing four decades (1982–2025) at level-2 areas is applied, enabling fine-grained regional forecasting in Yemen.
- The SMART-Rain framework incorporates lag features, rolling rainfall aggregations, anomaly indicators, cyclical temporal encoding, and logarithmic transformation to better represent complex rainfall dynamics.

- We use three sequence forecasting models (i.e., LSTM, Transformer Encoder, and iTransformer) under a unified experimental setting with Bayesian hyperparameter optimization. The results demonstrate that the iTransformer achieves the outperformance with an R^2 score of 0.7619, outperforming both LSTM and standard Transformer models.

A. Paper Organization

The rest of paper is organized as follows. Section II reviews the related work on rainfall forecasting. Section III presents the proposed methodology, including data preprocessing, feature engineering, and model architectures. Section IV describes the results and performance comparison of different models. Finally, Section VI concludes the paper.

II. RELATED WORK

Rainfall forecasting has long been recognized as a challenging time-series problem because precipitation exhibits strong nonlinearity, intermittency, seasonality, and extreme-event imbalance. Recently, data-driven models have increasingly complemented traditional statistical and numerical approaches, especially when long historical records and satellite-derived products are available. The CHIRPS dataset has become an important resource because it provides a long-period, high-resolution rainfall record from 1981 onward by blending satellite observations with station information [4]. This makes CHIRPS suitable for regional and subnational rainfall modeling in data-scarce environments.

Early deep learning studies on rainfall forecasting largely relied on recurrent architectures, especially LSTM, due to their ability to model sequential dependencies [6]. The authors of [7] proposed an LSTM-CP hybrid model for precipitation forecasting and showed that augmenting standard LSTM with a Chebyshev-polynomial-based module can improve prediction accuracy compared with ARIMA, SVR, and MLP baselines. Similarly, The authors of [8] applied CNN, GRU, and LSTM models to extreme rainfall prediction in Rwanda using multi-source climate data, including CHIRPS and ERA5, and reported that LSTM performed best among the tested recurrent models. These studies confirmed the usefulness of LSTM-based learning for rainfall dynamics; however, they are limited to station-based settings, classification of extremes, or relatively localized forecasting scenarios.

More recent research has explored Transformer-based models to better capture long-range dependencies and complex spatio-temporal interactions. The original Transformer architecture introduced self-attention as an alternative to recurrence for sequence modeling [12], [17]. Building on this idea, the authors of [14] proposed a transformer-based generative model for precipitation nowcasting using radar reflectivity data and showed superiority over several strong baselines, including cGANs, U-Net, ConvLSTM, and pySTEPS. In [15], the authors developed a transformer-enhanced spatio-temporal model, TransLSTMUNet, for precipitation forecast post-processing, combining convolution, Transformer blocks,

and ConvLSTM, and introduced a quantile-weighted loss to better address precipitation imbalance. These work demonstrated the promise of attention-based architectures, when spatial context or radar grids are available, but they targeted nowcasting weather prediction post-processing rather than long-term subnational rainfall forecasting.

A further development in multivariate time-series forecasting is the iTransformer, which reformulates the Transformer by treating variates as tokens rather than time points, allowing attention to model inter-variable correlations more directly [16]. The iTransformer was introduced as a general multivariate forecasting backbone and reported state-of-the-art performance on challenging real-world benchmarks [16], [18]. Although not originally designed for hydrometeorological rainfall applications, its variate-centric formulation is highly relevant for multivariate rainfall forecasting tasks involving multiple lagged, aggregated, anomaly, and seasonal features, such as those used in this study.

Another emerging direction is collaborative and geographically aware rainfall learning. The authors of [19] proposed collaborative station learning for rainfall forecasting, highlighting the value of learning from multiple stations instead of treating each location independently. This direction is relevant for regional rainfall prediction because nearby or climatologically related regions may share correlated rainfall behavior. However, many existing studies still do not encode subnational location identities through learnable embeddings. Compared with the above literature, the present study makes targeted contribution for long-term subnational rainfall forecasting over Yemen.

III. METHODOLOGY

This section presents the SMART-Rain methodology for multivariate rainfall forecasting. The SMART-Rain framework integrates data preprocessing, spatio-temporal feature engineering, sequence learning, and Bayesian hyperparameter optimization to model the nonlinear and variable rainfall patterns observed in long-term climate data.

A. Problem Formulation

The rainfall observations be represented as a multivariate time series:

$$\mathbf{X}_t = [x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(n)}] \in \mathbb{R}^n \quad (1)$$

where $\mathbf{X}_t$ denotes the feature vector at time step t , containing rainfall indicators, anomaly measures, rolling aggregations, and temporal variables. Given a historical sequence of length T :

$$\mathcal{S}_t = \{\mathbf{X}_{t-T+1}, \mathbf{X}_{t-T+2}, \dots, \mathbf{X}_t\} \quad (2)$$

The objective is to forecast the next rainfall value:

$$\hat{y}_{t+1} = f(\mathcal{S}_t; \theta) \quad (3)$$

where $f(\cdot)$ denotes the forecasting model parameterized by θ .

TABLE I
COMPARISON OF EXISTING STUDIES AND SMART-RAIN FRAMEWORK

Study	Data	Model	Forecast	Strength	Limitation relative to this work
[7]	City-level precipitation data, Yibin City, China	LSTM-CP hybrid	Precipitation forecasting	Hybrid recurrent architecture improves over ARIMA, SVR, and MLP	Focused on a localized setting; no subnational embedding or Transformer-based comparison
[8]	Multi-source climate data including CHIRPS and ERA5, four stations in Rwanda	CNN, LSTM, GRU	Extreme rainfall prediction	Compares multiple deep models and uses climate variables	Primarily station-based and event-focused; not long-horizon multivariate forecasting over many subnational regions
[14]	Radar reflectivity data, South Korea	Transformer-based generative model	Precipitation nowcasting	Strong Transformer-based nowcasting with transfer learning	Radar-image nowcasting task differs from dekadal subnational rainfall forecasting
[15]	TIGGE/ECMWF precipitation forecasts, southeast China	TransLSTMUNet + QWMSE	Forecast post-processing	Combines CNN, Transformer, and ConvLSTM; addresses imbalance	Targets NWP post-processing, not direct long historical rainfall forecasting with regional embeddings
[19]	Multi-station rainfall forecasting	Collaborative station learning	Rainfall forecasting	Exploits inter-station collaboration	Does not focus on CHIRPS-derived dekadal indicators or direct LSTM/Transformer/iTransformer benchmarking
SMART-Rain	CHIRPS-derived dekadal rainfall indicators, Yemen administrative level 2, 1982–2025	LSTM, Transformer Encoder, iTransformer with PCODE embedding	Multivariate long-horizon rainfall forecasting	Long historical span, subnational granularity, cyclical and lag features, learnable regional embedding, and direct recurrent vs. Transformer comparison	Still challenged by extreme rainfall underestimation; additional imbalance-aware losses may further improve results

B. Data preprocessing and feature engineering

To improve predictive performance, multiple feature engineering strategies were adopted.

1) *Lag Features*: Temporal dependencies were captured using historical lag variables:

$$x_t^{lag(k)} = x_{t-k}, \quad k \in \{1, 2, 3\} \quad (4)$$

2) *Cyclical Temporal Encoding*: To preserve seasonality, month and dekadal time indices were transformed into sine and cosine components:

$$month_{sin} = \sin\left(\frac{2\pi m}{12}\right), \quad month_{cos} = \cos\left(\frac{2\pi m}{12}\right) \quad (5)$$

$$dekad_{sin} = \sin\left(\frac{2\pi d}{36}\right), \quad dekad_{cos} = \cos\left(\frac{2\pi d}{36}\right) \quad (6)$$

where m is the month index and d is the dekadal index.

3) *Normalization*: All numerical variables were normalized using Min-Max scaling:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (7)$$

4) *Log Transformation*: Rainfall distributions are skewed, logarithmic transformation was applied to rainfall-related variables:

$$x_{log} = \log(1 + x) \quad (8)$$

This step reduces the influence of extreme values and improves optimization stability.

5) *Spatial Embedding*: To model regional heterogeneity, each administrative region (PCODE) was encoded into a learnable dense vector:

$$\mathbf{e}_i = \text{Embedding}(\text{PCODE}_i) \quad (9)$$

C. Sequence Construction

A sliding-window mechanism was employed to convert the time series into supervised learning samples. For each time

step, the previous T observations were used to predict the next rainfall value.

D. Deep Learning Models

Three forecasting architectures were evaluated.

1) *LSTM Model*: The LSTM network models temporal dependencies using gated memory cells:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (10)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (11)$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (12)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (13)$$

$$h_t = o_t \odot \tanh(C_t) \quad (14)$$

where f_t , i_t , and o_t are the forget, input, and output gates.

2) *Transformer Encoder*: The Transformer Encoder employs self-attention to model long-range temporal dependencies:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (15)$$

Multi-head attention is defined as:

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (16)$$

The encoder output is computed as:

$$\mathbf{Z} = \text{LayerNorm}(X + \text{MHA}(X)) \quad (17)$$

$$\mathbf{H} = \text{LayerNorm}(\mathbf{Z} + \text{FFN}(\mathbf{Z})) \quad (18)$$

3) *iTransformer model*: The iTransformer reformulates the Transformer architecture by treating variables as tokens rather than time steps, enabling stronger modeling of inter-feature dependencies. This design is highly suitable for multivariate rainfall forecasting, where lagged rainfall, anomaly indicators, and seasonal features interact strongly.

The final prediction is generated as:

$$\hat{y}_{t+1} = W_o \mathbf{H}_t + b_o \quad (19)$$

E. Loss Function

All models were trained by minimizing Mean Squared Error (MSE):

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (20)$$

Algorithm 1 SMART-Rain Forecasting Procedure

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1: Load rainfall dataset
2: Apply filtering and preprocessing
3: Perform normalization and log transformation
4: Generate lag and cyclical features
5: Encode PCODE regions using embeddings
6: Construct sliding-window sequences
7: for each model in {LSTM, Transformer, iTransformer}
   do
8:   Initialize parameters
9:   Optimize hyperparameters using Optuna
10:  for each epoch do
11:    Forward propagation
12:    Compute MSE loss
13:    Backpropagation
14:    Update parameters
15:  end for
16: end for
17: Evaluate using RMSE, MAE, and  $R^2$ 
18: Select best-performing model

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F. Hyperparameter Optimization

To ensure fair comparison, Bayesian optimization using Optuna was employed. The optimal parameter set was obtained by minimizing validation loss:

$$\theta^* = \arg \min_{\theta} \mathcal{L}_{val}(f(\theta)) \quad (21)$$

The search space include number of layers, hidden dimensions learning rate, dropout rate, batch size, and embedding dimension.

G. Training

Each model was trained using backpropagation with the Adam optimizer. Early stopping and validation monitoring were used to reduce overfitting. The same train-validation-test protocol was adopted for all models.

H. Evaluation Metrics

The forecasting models were evaluated using three standard regression metrics.

1) *Mean Absolute Error*:

$$\text{MAE} = \frac{1}{N} \sum |y_i - \hat{y}_i| \quad (22)$$

2) *Root Mean Squared Error*:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum (y_i - \hat{y}_i)^2} \quad (23)$$

3) *Coefficient of Determination*:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (24)$$

A higher R^2 and lower RMSE/MAE indicate better forecasting performance.

TABLE II
PERFORMANCE COMPARISON OF RAINFALL PREDICTION MODELS

Model	RMSE	MAE	R^2
LSTM	0.0611	0.0252	0.4574
Transformer Encoder	0.0924	0.0675	0.7251
iTransformer	0.0901	0.0635	0.7619

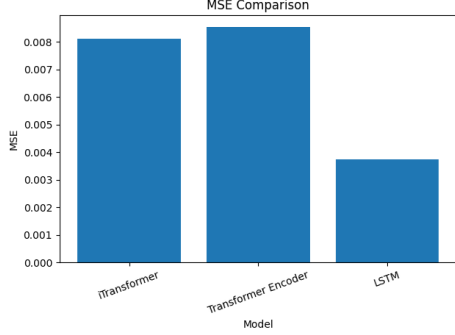


Fig. 1. Comparison of model performance in terms of Mean Squared Error (MSE).

IV. RESULTS AND DISCUSSION

This section presents the results obtained by applying three deep learning models (i.e., LSTM, Transformer Encoder, and iTransformer) to the Yemen rainfall dataset. Model performance is assessed using standard regression metrics, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). In addition, comparative visualizations and time-series prediction plots are used to further analyze forecasting behavior and model robustness.

A. Model performance comparison

Three models were evaluated for multivariate rainfall forecasting under the same preprocessing and training conditions. Table II summarizes the quantitative results.

The results show that the iTransformer achieves the best performance, obtaining the highest R^2 score of 0.7619, indicating the ability to explain rainfall variability and capture temporal dependencies compared with the other models. The Transformer Encoder performs competitively with an R^2 of 0.7251 outperforming the LSTM model. Although the LSTM model reports lower RMSE and MAE values, its relatively poor R^2 score (0.4574) suggests limited generalization capability and weaker ability to model complex nonlinear rainfall dynamics. Therefore, R^2 provides a more reliable indicator of forecasting effectiveness in this study.

B. Analysis of LSTM model

With a R^2 score of just 0.4574, the LSTM model shows modest predictive power, suggesting that it captures some short-term sequential relationships. Long-range temporal patterns and intricate nonlinear rainfall changes are difficult for

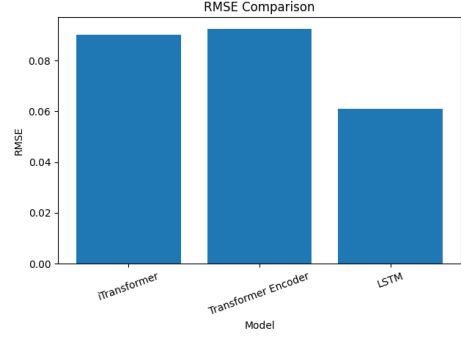


Fig. 2. Comparison of model performance in terms of RMSE.

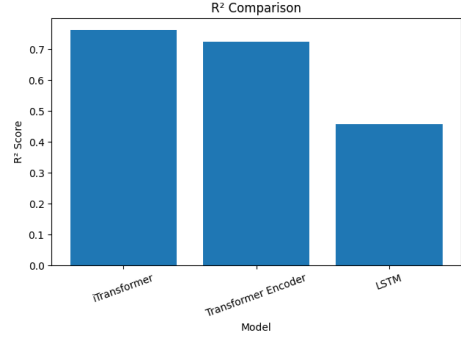


Fig. 3. Comparison of model performance in terms of (R^2).

LSTM to capture. The relatively low explanatory power suggests that recurrent architectures may be insufficient for highly variable rainfall data with strong seasonality and irregular extremes. This behavior is consistent with known limitations of LSTM models in handling long historical sequences and complex multivariate dependencies.

C. Performance of Transformer Encoder

The Transformer Encoder significantly improves forecasting performance over LSTM, achieving an R^2 score of 0.7251. This improvement highlights the effectiveness of self-attention mechanisms in learning long-range temporal relationships.

The model captures the general rainfall trend more effectively than LSTM and demonstrates better robustness to seasonal variations. However, slight deviations remain in extreme rainfall events, indicating that although attention-based models improve forecasting quality, highly skewed rainfall peaks remain challenging.

D. Performance of iTransformer Model

Among all tested models, the iTransformer provides the best performance with an R^2 score of 0.7619, RMSE of 0.0901, and MAE of 0.0635. These results confirm that the inverted Transformer architecture is highly effective for multivariate rainfall forecasting. The performance can be attributed to its enhanced ability to model inter-feature relationships while

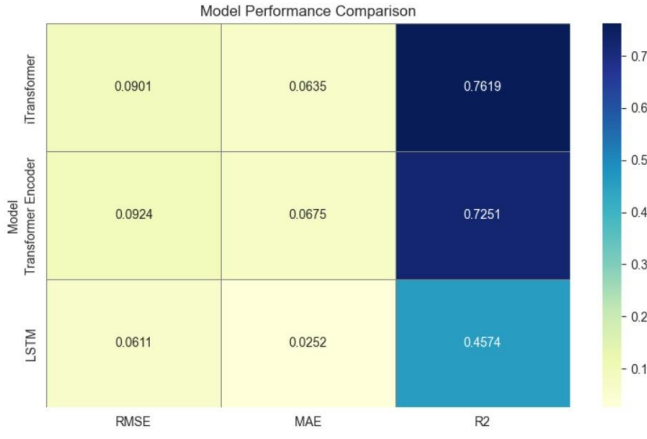


Fig. 4. Comparative model performance across evaluation metrics.

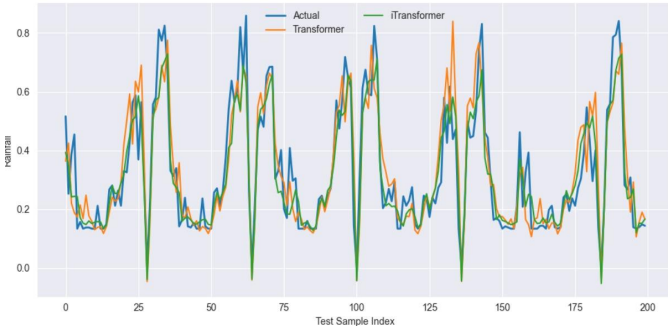


Fig. 5. Actual versus predicted rainfall.

preserving temporal structure. This is particularly important for rainfall forecasting, where lagged rainfall indicators, rolling averages, anomalies, and cyclical variables interact in complex ways.

Furthermore, the iTransformer demonstrates more stable predictions and better adaptation to sudden rainfall changes compared with the standard Transformer model. The actual versus predicted rainfall comparison in Fig. 5 further confirms that all models capture the overall rainfall pattern. However, the iTransformer follows the actual trend more closely during peak rainfall and sudden drops, demonstrating stronger temporal learning capability than both LSTM and the standard Transformer Encoder.

E. Impact of Log Transformation

Rainfall-related variables undergo transformation to lessen skewness and stabilize the training process. The pretreatment phase decreased the dominance of extreme values and enhanced numerical stability. Faster convergence and more balanced learning behavior were facilitated by the altered input characteristics, especially for low-to-moderate rainfall measurements. Nevertheless, extreme rainfall forecasting remains difficult due to the rarity and irregularity of such events.

F. Discussion

The findings show that for long-term rainfall forecasting, transformer-based designs perform better than recurrent neural networks. iTransformer strikes the optimal balance between resilience, generalization ability, and prediction accuracy. Long-range dependencies and inter-variable interactions are for rainfall prediction, as evidenced by the better performance of attention-based models. On the other hand, LSTM's worse performance implies that complicated climatic time-series data may require more than just recurrent structures. Despite these improvements, all models show reduced accuracy during extreme rainfall periods. This limitation is mainly caused by data imbalance, where high-intensity rainfall events occur much less frequently than normal rainfall values.

V. CONCLUSION

This paper presented SMART-Rain, a deep learning framework for multivariate rainfall forecasting using long-term CHIRPS-derived data at subnational levels. By leveraging more than four decades of rainfall observations (1982–2025), the proposed approach integrates spatio-temporal feature engineering, including lag features, cyclical encoding, anomaly indicators, and spatial embeddings, to model complex rainfall dynamics. Three sequence forecasting models, namely LSTM, Transformer Encoder, and iTransformer were evaluated. The results showed that iTransformer achieved the best performance with an R^2 score of 0.76, demonstrating strong capability in capturing long-range dependencies and multivariate interactions. Although the SMART-Rain learned temporal trends and seasonal variability, predicting extreme rainfall events remains challenging due to skewed and imbalanced rainfall distributions. SMART-Rain provides a scalable solution for rainfall prediction in data-scarce and geographically diverse regions. Future work will focus on improving extreme rainfall forecasting using weighted loss functions, quantile learning, hybrid models, external climate variables, and explainable AI techniques.

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